

A Real-Time Hardware-in-the-Loop Water SCADA Testbed for Cyber–Physical Research

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Abstract—Water facilities increasingly rely on Supervisory Control and Data Acquisition (SCADA) systems for real-time monitoring and control, making them critical targets for cyber–physical attacks. However, the lack of realistic, reproducible experimental platforms limits rigorous evaluation of control performance and cybersecurity mechanisms. This paper presents the design, implementation, and evaluation of a real-time hardware-in-the-loop (HIL) water SCADA testbed for cyber–physical research. The proposed testbed integrates industrial control hardware, a human–machine interface, and a high-fidelity virtual water process to emulate realistic water-treatment operations. A synchronized Python-based data-logging framework collects control, process, and network measurements in real time to support quantitative analysis and dataset generation. Experimental results demonstrate stable closed-loop water-level control with a steady-state error of 1.8 cm, 3.4% overshoot, and an average Modbus TCP latency of 50 ms (± 10 ms). The testbed provides a portable and cost-effective platform for studying water SCADA behavior and establishing baselines for future cybersecurity experimentation.

Index Terms—SCADA, Cyber-Physical System, Water Testbed, ModbusTCP, Siemens PLC, Factory I/O, Cybersecurity, Industrial Control Systems(ICS), Operational Technology(OT).

I. INTRODUCTION

Supervisory Control and Data Acquisition (SCADA) systems are fundamental to industrial automation, providing real-time monitoring and control across critical sectors such as water treatment, manufacturing, energy distribution, and oil and gas. Their reliability and availability are essential for national infrastructure safety. However, the increased interconnectivity of SCADA networks driven by modernization, Industry 4.0 integration, and IoT adoption has exposed these systems to unprecedented levels of cyber risk.

Modbus, originally introduced by Modicon in 1979 [1], remains one of the most widely deployed industrial communication protocols due to its simplicity, openness, and vendor-neutral design. Its Ethernet-based variant, Modbus TCP, has become the dominant communication standard in contemporary SCADA systems. Modbus TCP supports high-speed polling, real-time sensor acquisition, and deterministic device-to-device communication, making it central to distributed industrial processes such as water pumping, chemical dos-

ing, distributed energy management, automated manufacturing lines, and intelligent vehicle subsystems.

However, Modbus TCP was not designed with security in mind. Its lack of built-in authentication, encryption, and integrity checks has made it a frequent target for cyber adversaries. Recent threat intelligence reports highlight increasing attacks on SCADA networks [2] [3] [4], including VOLTZITE [5] operations targeting U.S. utility providers and the FrostyGoop [6] Industrial Control System (ICS) malware that disrupted heating infrastructure in Ukraine. These incidents underscore the urgency of developing safe, realistic, and modular SCADA testbeds for security experimentation, attack simulation, and machine-learning-based intrusion detection.

Despite existing work in the ICS cybersecurity domain, researchers still lack reproducible, cost-effective testbeds that combine real industrial hardware with high-fidelity virtual process simulation. Many published works rely on restricted datasets, unavailable hardware, or proprietary industrial software.

The objective of this work is to develop a real-time SCADA testbed for critical water infrastructure by designing and implementing a closed-loop water-level control system. The testbed enables safe replication of industrial control system behavior and establishes baseline cyber–physical dynamics to support future security experiments and anomaly detection. The Siemens S7-1200 PLC was selected due to its widespread industrial adoption, support for standard protocols such as Profinet and Modbus TCP, cost-effectiveness, and suitability for reproducible testbed deployments.

The following are the significant contributions:

- A reproducible hardware-in-the-loop SCADA water-level testbed integrating a Siemens S7-1200 PLC, Weintek HMI, and Factory I/O via Modbus TCP.
- A synchronized Python-based logging and evaluation framework that captures PLC and process data in real time, quantitatively assesses system latency, stability, and control performance.

The rest of the paper is structured as follows: Section II reviews related work; Section III describes the system architecture; Section IV details the implementation; Section V presents an experimental evaluation; Section VI concludes

the paper findings and talks about the limitation; and future works

II. RELATED WORK

In this section, we provide background on SCADA systems, Modbus TCP, and the use of Siemens S7-1200 in industrial automation. The increasing exposure of ICS and SCADA networks to cyber threats has led to significant research into securing communications, identify vulnerabilities, and protecting legacy systems without sacrificing availability.

A. Security Vulnerabilities of SCADA

Security studies have examined vulnerabilities in ICS networks as adversaries increasingly target operational technology. AL Ghazo and Kumar [7] introduced ANDVI, a passive fingerprinting system capable of identifying SCADA devices and mapping them to known vulnerabilities without disturbing sensitive equipment. Carson et al. [8] examined SCADA weaknesses and recommended layered defenses including encryption, IDS, and network segmentation. They highlight the risks posed by IT/OT convergence, citing historical incidents such as Stuxnet and Ukraine's power grid attacks.

B. S7-1200 Programmable Logic Controller (PLC)

The Siemens S7-1200 family of PLCs, including the CPU 1212C model used in this research, has been adopted in both industry and academia due to its compact design, integrated Profinet communication and expandability, making it suitable for applications ranging from motor control to IoT and secure communication. Prior studies highlight its versatility: Salkić et al. demonstrated its robustness in DC motor control with open and closed-loop PID configurations through TIA Portal and HMI interfaces [9] Saha et al. employed the S7-1200 in an IoT-enabled pharmaceutical stability chamber, integrating WinCC Runtime Advanced for real-time monitoring and FDA-compliant data handling; [10] Dao et al. evaluated its role in secure Modbus TCP/IP communications, exposing susceptibility to ARP spoofing attacks and proposing hardware-based mitigations; [11] and Finck and Dohrmann reverse-engineered the S7CommPlus protocol, revealing cryptographic weaknesses and legacy vulnerabilities that persist even after Stuxnet [12]. Although these works apply the PLC to automation, IoT, and security contexts, our research uniquely employs the S7-1200 CPU 1212C in a critical water infrastructure testbed, extending its application toward cyber-physical resilience and infrastructure protection.

C. Crypto-Based Solutions to Protect ModBus TCP

Modbus TCP, lacking encryption and authentication, is vulnerable to replay, injection, modification, and man-in-the-middle attacks. Xuan and Yongzhong [13] proposed Modbus-S, an enhanced protocol integrating encryption and signatures. Katulić et al [14].designed a MAC-based protection scheme using Chaskey-12, enabling secure communications even with legacy devices. Our testbed will serve as a foundation for future research that will focus on the security for water infrastructure using Modbus TCP

D. FACTORY I/O As Virtual ICS Process Simulator

Factory I/O is used in much of the existing literature as a high-fidelity virtual plant for PLC programming and control education, where it enables safe experimentation with realistic industrial processes without the risks associated with physical equipment [15], [16]. Beyond instructional use, it has also been adopted in automation research as a digital factory environment for modeling and monitoring PLC-based production systems [17], as well as in hardware-in-the-loop (HIL) configurations where the virtual process is controlled in closed loop by a real PLC [18].

Only limited prior work extends Factory I/O into OT/ICS security research, primarily using virtualized processes to evaluate defensive mechanisms without risking real infrastructure [19]. In contrast, this work integrates Factory I/O as a virtual water-tank process within a hardware-in-the-loop SCADA testbed built around a real Siemens S7-1200 PLC, enabling repeatable cyber-physical experimentation and synchronized process-level data logging. Unlike many existing testbeds that rely on proprietary platforms or lack coordinated control and process data collection, the proposed system provides a reproducible and extensible environment tailored to water-infrastructure security research.

III. SYSTEM ARCHITECTURE

A. SCADA Testbed Design

The water-level testbed is designed as a modular, hardware-in-the-loop (HIL) platform that integrates a physical PLC, a graphical HMI, a virtual industrial process, and a data-logging engine as shown in Figure.1. The experimental workflow replicates the control and communication workflow found in real-world water treatment systems, while maintaining full experimentation and cybersecurity evaluation.

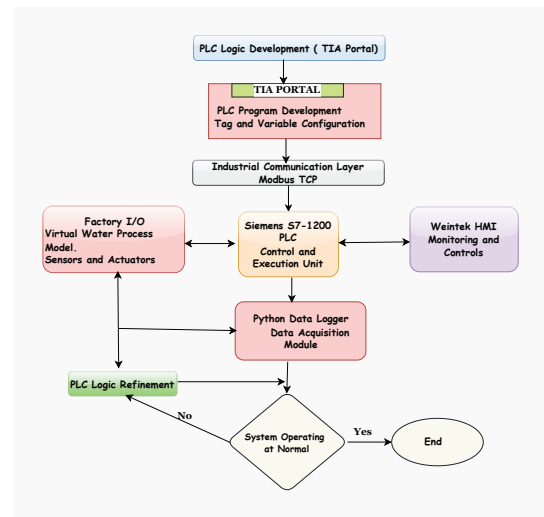


Fig. 1. Experimental Workflow of the SCADA Testbed

Figure.2 illustrates the end-to-end data flow within the proposed SCADA testbed. Process measurements generated by the virtual water tank in Factory I/O are transmitted to the

PLC via Modbus TCP, where control logic computes actuation commands. These commands are sent back to the virtual process to regulate inflow and outflow. Operator inputs from the HMI are written to the PLC control database, while internal PLC states and network transactions are passively collected by the Python data logger, enabling synchronized observation of control, process, and communication behavior.

The S7-1200 PLC operates as the core controller, executing level-regulation logic and exchanging real-time process variables with a simulated water tank in Factory I/O. A Weintek MT8070iE HMI provides operator-level interaction, including setpoint adjustments, mode selection, and process visualization. A Python-based logger captures synchronized PLC and process data streams, enabling high-resolution cyber-physical dataset generation. This design ensures that deterministic process behavior, controlled experimentation environments, seamless integration of cyber-attacks, network anomalies, and machine-learning-based intrusion detection in future stages.

B. System Components and Network Workflow

Here, we discuss the system components required for the proposed hardware-in-the-loop SCADA testbed, and highlight the network workflow for the integration of these components. For our research, the system components used are; a Siemens S7-1200 PLC, a Factory I/O virtual water-tank process, a Weintek HMI panel, and a Python-based data-logging module. Together, these components form a cohesive environment that supports realistic control execution, operator interaction, and cybersecurity experimentation.

All components communicate over a dedicated Local Area Network(LAN) using standard industrial protocols. The PLC serves as the central control node, exchanging process data with Factory I/O via Modbus TCP, while the HMI communicates with the PLC through S7Comm for operator interaction. A Python-based logger connects passively to both interfaces using Snap7 and Modbus libraries to collect synchronized control and process data without affecting real-time operation.

Devices are interconnected through an unmanaged ethernet switch and assigned static IP addresses to ensure deterministic communication. By isolating the testbed on a private subnet, the architecture minimizes external traffic interference and ensures repeatable experimentation, closely reflecting industrial deployments where control systems operate on dedicated operational networks.

IV. WATER SCADA TESTBED IMPLEMENTATION

The proposed testbed combines physical industrial hardware with a virtualized process environment. The Siemens S7-1200 PLC and Weintek HMI are real industrial devices, while the water tank, sensors, and actuators are emulated within Factory I/O. This hybrid configuration enables safe experimentation without physical water infrastructure while preserving realistic control behavior.

In Fig. 2, we present the implementation of our proposed water SCADA testbed, illustrating how PLC, industrial communication, virtual process simulation, operator interaction,

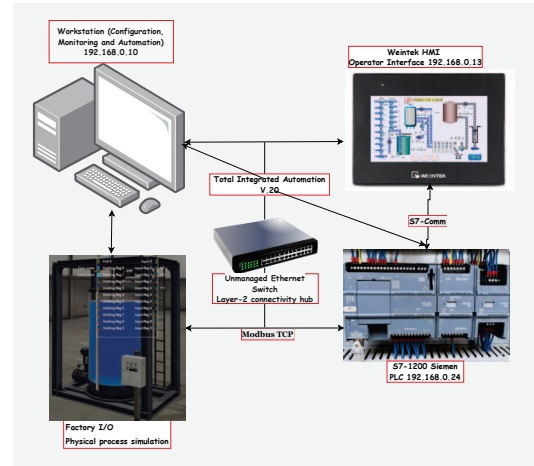


Fig. 2. Physical and logical network architecture of the water SCADA testbed, illustrating device interconnections, protocol usage, and IP-layer isolation during experiments.

and data logging are integrated during runtime. It illustrates the closed-loop interaction between the devices, as well as the iterative refinement process used to validate stable system operation before experimental data collection.

Table I summarizes the core testbed components including the PLC, HMI, virtual process, data-logging module, and host workstation along with their corresponding hardware and software configurations.

TABLE I
TESTBED COMPONENTS AND HARDWARE CONFIGURATION

Device	Description	Configuration
PLC	Siemens S7-1200	1212C DC/DC/DC
HMI	Weintek MT8070iE	eMT3070 Series (17121000)
Virtual Tank	Factory I/O	Ultimate Edition (v2.5.10)
Python Logger	Snap7 + Modbus TCP	Python 3.12
MSI Katana 13thGen	Windows 11 Pro	NVIDIA RTX 4050GPU

The Siemens S7-1200 CPU 1212C, serves as the core automation controller for the water-level process. Its built-in Profinet Ethernet interface enables deterministic communication with both the HMI and the virtual process, supporting reliable closed-loop control under real-time operating conditions.

A. Programmable Logic Controller (PLC)

The Siemens S7-1200 CPU 1212C, serves as the core automation controller for the water-level process. Its built-in Profinet Ethernet interface enables deterministic communication with both the HMI and the virtual process, supporting reliable closed-loop control under real-time operating conditions.

During operation, the PLC executes cyclic control logic that integrates process acquisition, control computation, and actuator commanding. Level and flow measurements are read from Factory I/O registers via Modbus TCP, then scaled and validated before being processed by the control logic.

A proportional–integral (PI) controller computes the control signal based on the error between the operator-defined setpoint and the measured water level.

The PI controller gains were selected through empirical tuning, starting from conservative values and iteratively adjusted to achieve stable response, minimal overshoot, and acceptable settling time under nominal operating conditions.

Computed control outputs are translated into inflow and outflow valve actuation commands, which are written as binary coil values to the Factory I/O simulation in real time. To support operator interaction, a dedicated data block (DB_HMI) is used to store HMI inputs, including the setpoint (SP_{cm}) and operating mode ($Mode_Auto$), while exposing internal process states such as the measured level (PV_{cm}) and control output (U_{pct}). This organization maintains a clear separation between control logic and operator interface, simplifying system validation and future extension.

B. Factory I/O Integration

Factory I/O provides a high-fidelity virtual representation of a water-tank process, including inflow and outflow valves, a flow meter, and an analog level sensor. The simulation operates as a Modbus TCP server, allowing the PLC and Python data logger to exchange process data in real time. Sensor measurements and actuator commands are mapped to Modbus registers and coils, enabling closed-loop interaction with the control logic. This virtualized approach captures essential water-level dynamics while eliminating the safety risks and operational constraints associated with physical tanks, ensuring repeatable behavior for control validation and dataset generation.

C. Human-Machine-Interface(HMI)

The Weintek MT8070iE HMI provides operator-level interaction with the testbed by displaying process variables and allowing setpoint and mode selection. Configured using EasyBuilder Pro, the panel communicates with the PLC via the Siemens S7-1200 driver and exchanges data through a dedicated HMI data block. By operating on the same subnet as the PLC, operator actions are reflected immediately in the control logic, enabling seamless closed-loop operation. The HMI serves as the human-facing layer of the testbed, analogous to supervisory panels used in industrial control rooms.

D. Integrated System Workflow and Data Logging

The proposed SCADA testbed functions as a unified cyber–physical system in which the PLC, HMI, Factory I/O virtual process, Python data logger, and engineering workstation interact through realistic industrial communication layers. This integration supports coordinated control execution, operator interaction, and continuous process monitoring within a single operational workflow. To enable experimental evaluation and security analysis, a Python-based data-logging module continuously collects synchronized data from both the PLC and the virtual plant. The logger connects to the PLC via Snap7 and to Factory I/O using Modbus TCP, captures operator

inputs, internal control states, and process variables, and records the data in a time-stamped format for offline analysis. A sampling interval of 200–500ms was selected to balance real-time observability with network stability, consistent with the slow dynamics of SCADA systems of water-treatment. This synchronized logging approach allows reliable correlation of process behavior, control actions, and network activity, providing a solid foundation for future attack analysis, timing-based anomaly detection, and intrusion-detection research. It does not introduce bias into latency measurements, as each Modbus transaction is independently timestamped at request and response boundaries.

V. EXPERIMENTAL RESULTS

A. Closed-Loop Control Performance

The testbed exhibits stable closed-loop water-level control under normal operating conditions. When operating in automatic mode, the PLC computes the control error between the measured level and the operator-defined setpoint and adjusts valve actuation accordingly. Experimental results show a rise time of approximately 2.5s, a settling time of 6s, limited overshoot (3.4%), and a steady-state error of 1.8cm. As illustrated in Fig.3, the tank-level response converges smoothly to the setpoint without oscillation, confirming that the virtual process dynamics, control logic, and sampling rate are well aligned and suitable for cyber–physical security experimentation.

To validate the closed-loop behavior of the water-level control system, a step change in setpoint was introduced while the testbed operated in Auto mode. The process variable (PV), setpoint (SP), and valve actuation signal were recorded through the Python data-logging engine to capture the real-time dynamics of the PLC–process interaction.

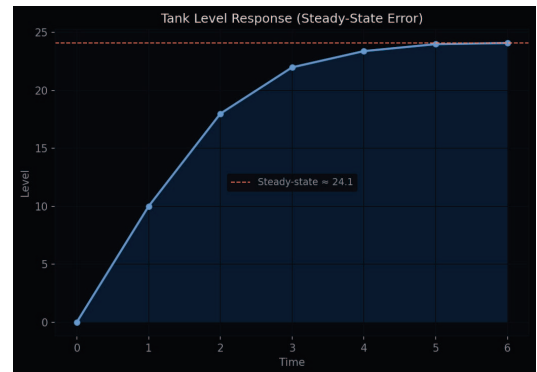


Fig. 3. Tank level response steady-state error

B. Network Performance Evaluation

Modbus TCP latency was measured as round-trip time (RTT) by timestamping each request at the Python data logger immediately before issuing a Modbus read to the Factory I/O server and timestamping again upon reception of the corresponding response. RTT samples were collected continuously over a 30-minute period during normal closed-loop operation to assess whether the testbed meets real-time

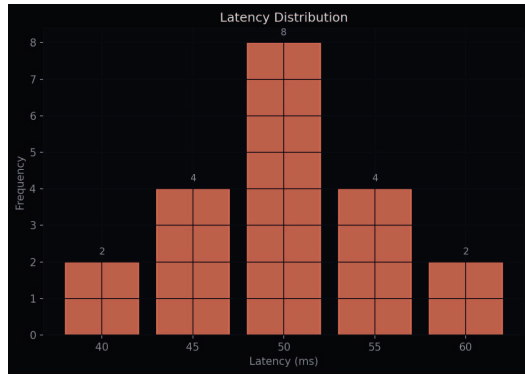


Fig. 4. Modbus TCP latency distribution

communication requirements for industrial control. The system exhibited a mean round-trip latency of approximately 50ms, with limited jitter (± 10 ms) and a maximum observed delay of 68ms, while no packet loss was recorded. As illustrated in Fig. 4, the tight clustering of latency values indicates stable and deterministic network behavior, confirming predictable process update intervals suitable for real-time water SCADA control and establishing a baseline for replay analysis, timing-based anomaly detection, and future cyberattack evaluation.

The round-trip time for the i th Modbus transaction is defined as

$$RTT_i = t_{\text{recv}}^{(i)} - t_{\text{send}}^{(i)}, \quad (1)$$

where $t_{\text{send}}^{(i)}$ and $t_{\text{recv}}^{(i)}$ denote the transmission and reception timestamps, respectively. The mean network latency is computed as

$$\overline{RTT} = \frac{1}{N} \sum_{i=1}^N RTT_i, \quad (2)$$

where N is the total number of sampled transactions.

VI. CONCLUSION

This paper presented the design and implementation of a real-time hardware-in-the-loop water SCADA testbed integrating a Siemens S7-1200 PLC, a Weintek HMI, and a Factory I/O virtual process. Experimental results demonstrate stable closed-loop control and reliable Modbus TCP communication, confirming the testbed's suitability for cyber physical experimentation. The synchronized Python-based data-logging framework enables high-quality dataset generation, providing a practical foundation for intrusion-detection research and security analysis in water-sector SCADA environments.

Although the testbed provides realistic SCADA functionality, its design reflects deliberate scoping choices to support safe and repeatable cybersecurity experimentation. The virtual water-tank process abstracts physical effects such as turbulence, component wear, and complex hydraulic dynamics, which are not essential for evaluating control behavior or network-level security interactions. The current implementation focuses on a single PLC and a simplified network

topology to prioritize depth of analysis, experimental repeatability, and clear interpretation of results. Encrypted industrial communication, such as TLS-enabled Modbus, is not yet enabled, allowing baseline system behavior to be established for subsequent security evaluation and extension. Future work will extend the testbed to generate dataset for future research and making it available for advancing Water Scada research. Physical process integration, including a real water tank and industrial sensors, will be explored to complement the virtual plant.

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